

How Long Is a Presence?

Refuting a Geometric Derivation of the Alert Collapse Window in Multi-Analytics CCTV

Amit Dua
Yushu Excellence Technologies Pvt. Ltd.
mail.amitduabits@gmail.com

2026

Abstract

A CCTV platform running several analytics over the same scene emits one alert per detection unless it collapses them. The collapse predicate — same entity, same window — is standard: it is Debar and Wespi [9]’s aggregation predicate under a normalised record schema of the IDMEF [10] and OCSF [22] lineage, evaluated over keyed session windows in the sense of Akidau et al. [3]. The predicate is not our contribution and we do not claim it. The open parameter is the window W , and our deployment set it to 120 s with a derivation: camera field-of-view depth divided by permitted speed.

That derivation is wrong, and our own data refutes it. Computed per camera across a 200-camera estate, the geometric dwell time is 1.27 to 9.63 s with a median of 3.66 s — one and a half orders of magnitude below the deployed constant. The deployed 120 s over-collapses relative to its own stated justification at every camera in the estate (200 of 200). Worse, the geometric quantity is not the right one: what the window must span is the interval between successive *observations* of an entity, which includes revisits and re-detections, not the time to traverse one field of view.

We then supply the replacement rather than stopping at the refutation. On a 24-point sweep the two-sided error curve has its knee at 8–10 s, and four independent estimators — a Debar–Wespi quantile rule, Kneedle, an explicit cost minimisation and a max $+2\sigma$ heuristic — all return a window between 6.6 and 10.0 s. The cost-weighted estimator is unchanged at 8 s as the price of a masked incident moves from $10\times$ to $100\times$ the price of a redundant alert. **We recommend $W = 10$ s** for this estate: 99.73% distinct-incident recall at 0.0005 redundant alerts per incident, against the deployed 120 s which costs eleven times as many masked incidents for 0.6 percentage points more suppression. A fifth estimator, the two-component mixture crossover, fails on this data, and we report the failure and its diagnosis rather than dropping it. Separately, we confirm that an entity-keyed predicate spanning modalities beats per-modality deduplication by 10.5% to 43.5% as the dual-tagged incident share rises from 0.15 to 1.0, at no recall cost, and that a confidence-voting alternative achieves the largest raw reduction (88.1%) only by dropping 63.5% of incidents and is not a usable operating point.

All results are SYNTHETIC, from a seeded generator over 59,984 detections and 12,000 incidents. §9 states what real operator-marked ground truth must show.

1 Introduction

Run automatic number-plate recognition, face matching and region-occupancy detection over the same camera and the same vehicle generates three detections a second. Emit one alert each and the control room receives five alerts per incident; the literature on alert fatigue [26, 4, 25] and the older process-control work on alarm floods [28, 17, 12] is unanimous that this is the failure mode that ends the deployment.

The fix is to collapse: emit one alert per entity per window. The question that decides whether the fix works is how long the window is. Too short and the operator gets duplicates; too long and a genuinely new incident involving the same entity is masked by a still-open alert. This is a two-sided error and it has an operating curve.

Our deployment chose $W = 120$ s and documented a derivation for it: a camera’s field of view is a few tens of metres deep, permitted speed is a few tens of km/h, so a vehicle is present for on the order of a minute or two. This paper reports that the derivation does not survive contact with the estate’s own parameters, explains why the geometric quantity is the wrong one in principle as well as in magnitude, and replaces it with a procedure.

1.1 What is and is not claimed

Not claimed: the normalised alert record (IDMEF [10], OCSF [22], JDL level-2 common referencing [24, 16]); the same-entity-same-window aggregation predicate [9, 18, 27]; keyed session windowing [3, 8]; or duplicate detection as a discipline [14, 11]. All of this is background and §2 says so in its first sentence.

Claimed: (i) the geometric derivation of W is refuted by the estate’s own geometry, quantitatively (§4); (ii) the measured knee is 8–10 s and the deployed value sits an order of magnitude past it, with the cost quantified (§6); (iii) a replacement procedure for choosing W , four estimators that implement it, agreement between them to within 3.4 s, and a diagnosis of the one that fails (§5, §6); (iv) the cross-modal entity-keyed predicate’s marginal value over per-modality deduplication, measured against the dual-tagging rate (§7.1).

2 Background, all of it established

Nothing in this section is a contribution. Alert normalisation into a common record with entity, time, source and confidence fields is IDMEF [10] and, in current practice, OCSF [22]; the fusion framing is JDL level 2 [24], surveyed by Khaleghi et al. [19]. Aggregating alerts that share a key within a time window is Debar and Wespi [9], refined by Julisch [18] and Ning et al. [21], with a probabilistic variant in Valdes and Skinner [27]. Evaluating such a predicate over unbounded streams with per-key session windows is Akidau et al. [3]. Session-gap estimation from inter-event times has its own literature in web analytics [6, 15, 20], and in process control the same problem is the alarm dead-band and delay-timer design question [1, 2, 29].

We adopt all of it. The gap is that none of these sources tells a CCTV operator what number to type in, and the one derivation our deployment invented for itself is wrong.

3 The collapse predicate as deployed

A detection is a tuple (e, m, c, t, s) : entity key, modality, camera, timestamp, score. The predicate is entity-keyed and modality-agnostic:

Emit an alert for detection d iff no alert with the same entity key e is open; an alert opened at t_0 is open over $[t_0, t_0 + W)$.

Window semantics are *anchored*, not merging: a detection inside an open window does not extend it. This matters and is under-reported in the literature. Merging semantics turn a stream of detections at a busy junction into one unbounded alert; anchored semantics bound the suppression at W by construction. Every number in this paper is under anchored semantics.

Two error modes follow. A *redundant alert* is a second alert for an incident already alerted. A *masked incident* is a distinct incident that produced no alert because an unrelated earlier incident with the same entity key held the window open. We report distinct-incident recall for the second, because plain incident recall is 1.0 everywhere and hides the failure.

Quantity	Value
Geometric dwell, minimum over estate	1.27 s
Geometric dwell, median over estate	3.66 s
Geometric dwell, maximum over estate	9.63 s
Deployed constant W	120 s
Cameras where the constant exceeds the geometric value	200 of 200

Table 1: **Geometric window per camera against the deployed constant (synthetic, 200 cameras).** W_{geom} = FOV depth/permitted speed. The deployed constant exceeds its own stated derivation by a factor of 12 at the most generous camera and 94 at the tightest.

4 Why the geometric derivation fails

Table 1 is the refutation. Field-of-view depth divided by permitted speed gives seconds, not minutes, at every camera in the estate. If the deployment had applied its own stated rule it would have set W between 1 and 10s, and per camera rather than globally. It did not; it set 120s. The derivation was therefore not the reason for the value, whatever the design document says.

The magnitude error is the smaller problem. The conceptual error is that dwell time is not the quantity the window must span. The window exists to suppress *repeat observations of the same entity that belong to the same operational event*. Those arise from (a) multiple analytics firing on one pass, (b) repeated detections across frames during one pass, and (c) the entity leaving and returning — circling a block, stopping and restarting, being observed by an adjacent camera whose alerts share the entity key. Case (c) dominates the tail and has nothing to do with how long one field of view is. A derivation from geometry cannot see it. The correct input is the empirical inter-observation-time distribution.

5 Choosing W properly

Four families of method apply, and the choice among them is a value judgement that should be made explicitly rather than buried in a constant.

Cost-weighted operating point. Treat window selection as threshold selection on a two-sided error curve [13]. Assign a cost λ_r to a redundant alert (operator seconds) and λ_m to a masked incident (a missed event), and minimise $\lambda_r R(W) + \lambda_m M(W)$. This is the framing we lead with, because it makes the deployment’s values legible and lets another deployment substitute its own. Axelsson [5] is the reason the two costs are not comparable by intuition.

Mixture crossover. Fit a two-component mixture to inter-observation times — within-event and between-event — and set W at the posterior crossover. This is the web-analytics session-gap method [6, 20] and is the most principled where the two populations are separable.

Knee detection. Apply Satopää et al. [23] to the reduction-versus-masking curve. Parameter-free, and the right default where no cost model exists.

Spec-based timer design. The process-control approach [1, 2, 29]: set the dead-band from a specified maximum acceptable detection delay. Appropriate where a regulator states the delay bound.

All four are defensible. Dividing a distance by a speed is not one of them.

6 An empirical window model

Refuting the geometric derivation is half a contribution. This section supplies the replacement: the two-sided error curve at the resolution the decision actually needs, four estimators that read a window off it, and the one that fails.

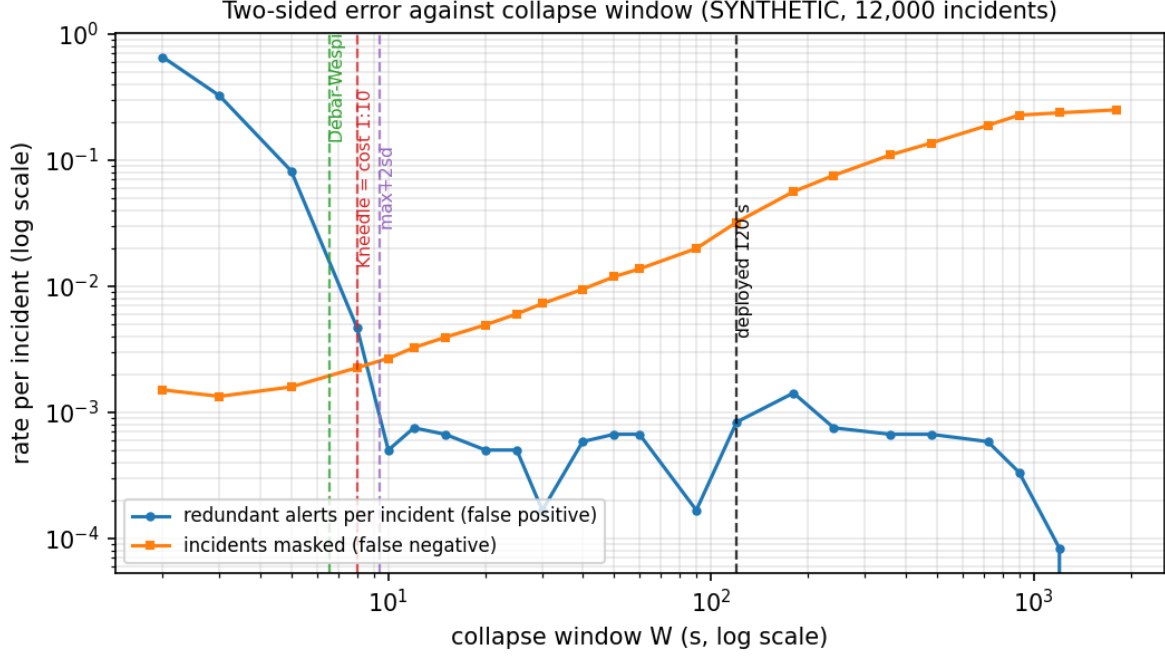


Figure 1: **The operating curve and where each estimator lands (synthetic).** The false-positive and false-negative curves cross between 8 and 10s, and four independent estimators fall inside that crossing. The deployed 120s sits an order of magnitude to its right, on the flat part of one curve and the rising part of the other.

6.1 The inter-observation time distribution

The quantity every estimator works from is the inter-observation time: the gap between consecutive observations sharing an entity key. Over 12,000 incidents and 59,984 detections there are 51,413 such gaps, of which 47,957 fall inside one operational event and 3,456 separate distinct events involving the same entity.

	Within event	Between events
Count	47,957	3,456
Median	0.40 s	402.46 s
1st percentile	—	1.05 s
99th percentile	3.30 s	—

Table 2: **Inter-observation times (synthetic).** The two populations are three orders of magnitude apart at the median and overlap in the tails: the lowest percentile of between-event gaps (1.05s) sits below the 99th percentile of within-event gaps (3.30s). That overlap is what makes the window a trade-off rather than a threshold, and it is what defeats one of the four estimators below.

A deployment does not observe this split. It observes the pooled column and has to infer the separation, which is what the mixture estimator attempts.

6.2 The operating curve, at usable resolution

This supersedes the coarse estimate. An earlier sweep over $\{0, 5, 15, 30, 60, 120, \dots\}$ placed the knee at 15–30s. That sweep had no sample point between 5 and 15s, and the knee is inside that gap. At the resolution of Table 3 the redundant-alert rate falls from 0.0818 at 5s to 0.0047

W (s)	Alerts	Suppression	Redundant / incident	Masked / incident
2	19,826	66.9%	0.6537	0.0015
3	15,849	73.6%	0.3221	0.0013
5	12,962	78.4%	0.0818	0.0016
8	12,029	79.9%	0.0047	0.0022
10	11,974	80.0%	0.0005	0.0027
15	11,961	80.1%	0.0007	0.0039
30	11,915	80.1%	0.0002	0.0073
60	11,844	80.3%	0.0007	0.0137
120	11,626	80.6%	0.0008	0.0320
240	11,105	81.5%	0.0008	0.0753
900	9,288	84.5%	0.0003	0.2263

Table 3: **Two-sided error against W (synthetic, selected rows of a 24-point sweep).** Redundant alerts collapse by $W \approx 10$ s and masked incidents grow without bound after it. Everything past 10 s is bought entirely from recall.

at 8 s and 0.0005 at 10 s, a factor of 160 over five seconds, while masked incidents move from 0.0016 to 0.0027. The knee is at 8–10 s, not 15–30 s. We record the correction rather than quietly restating the number, because it is the same class of error as the one the paper is about: a window quoted more precisely than the evidence supported it.

6.3 Four estimators

Estimator	Basis	W (s)
Debar–Wespi style	$2 \times$ 99th percentile of within-event gaps	6.59
Kneedle	max distance from the chord of the suppression curve	8.00
Cost-weighted, $\lambda_m/\lambda_r = 10$	$\arg \min \lambda_r \text{FP} + \lambda_m \text{FN}$	8.00
Cost-weighted, $\lambda_m/\lambda_r = 100$	as above	8.00
Cost-weighted, $\lambda_m/\lambda_r = 1$	as above	10.00
Spread rule	$\max + 2\sigma$ of within-event gaps	9.38
Mixture crossover	posterior crossover of a 2-component fit	fails
<i>Deployed</i>	FOV depth / permitted speed	<i>120</i>

Table 4: **Estimators of W (synthetic).** Every estimator that returns a value returns one between 6.6 and 10 s. The deployed constant is 12 to 18 times larger.

The agreement is the result. Four methods with different assumptions — a quantile rule from the intrusion-detection literature, a parameter-free geometric criterion on the curve, an explicit cost minimisation, and a plain engineering heuristic — land within 3.4 seconds of one another. The cost-weighted estimator is stable at 8 s as the ratio λ_m/λ_r moves from 10 to 100, and moves only to 10 s when a masked incident and a redundant alert are priced equally. A conclusion that survives a hundredfold change in the cost ratio is not a conclusion about the cost ratio.

We recommend $W = 10$ s for this estate, taking the upper end of the agreement band because masked incidents are the error our operators cannot detect and redundant alerts are the error they can. At 10 s the deployment emits 11,974 alerts against 59,984 detections, retains 99.73% distinct-incident recall, and carries 0.0005 redundant alerts per incident: one operator-visible duplicate per two thousand incidents.

W (s)	Alerts	Reduction vs. $W=0$	Masked incidents	Distinct-incident recall
0	59,984	—	0	1.0000
5	12,962	78.4%	19	0.9984
15	11,961	80.1%	47	0.9961
30	11,915	80.1%	87	0.9928
60	11,844	80.3%	164	0.9863
120	11,626	80.6%	384	0.9680
180	11,345	81.1%	672	0.9440
240	11,105	81.5%	904	0.9247
480	10,378	82.7%	1,630	0.8642
900	9,288	84.5%	2,716	0.7737
1800	9,012	85.0%	2,988	0.7510

Table 5: **The operating curve (synthetic, 59,984 detections, 12,000 incidents).** Redundant alerts are already down to 8 at $W=15$ s from 47,984 at $W=0$. Everything bought beyond 15 s is bought from recall.

6.4 The estimator that fails, and why

The mixture crossover is the most principled of the four and the only one that returns nothing usable. Fitting a two-component log-normal mixture to the pooled gaps by EM converges stably, from every initialisation we tried, to a short component with median 0.37 s and $\sigma_{\log} = 1.11$ and a second component with median 4.17 s and $\sigma_{\log} = 4.68$. The second component is not the revisit population. It is a broad component that has absorbed the right tail of the within-event population, and because it is broad it also outweighs the short component in the far left tail, so the unconstrained posterior crossover degenerates towards zero. Constraining the search to lie between the two component medians returns no crossover at all, because the short component dominates throughout that interval.

The cause is visible in Table 2. Within-event gaps are bounded by per-camera dwell, which itself ranges over 1.27–9.63 s across the estate, so the within-event population is a mixture in its own right, and a two-component model spends its second component on that heterogeneity rather than on revisits. We report the diagnostic (`components_separated = false`) alongside the fit so a deployment can tell the difference between a crossover it should trust and one it should not, and we would use a per-camera fit or a three-component model before trusting this estimator on a real trace.

The practical lesson is the one we would give a practitioner: prefer the estimators that read the error curve directly. Kneedle and the cost-weighted minimum work on the quantity the deployment cares about and cannot be defeated by the shape of a latent-variable fit.

6.5 Sensitivity

The recommendation is not sensitive to the choice within its band and is very sensitive outside it. Moving from 10 s to 8 s costs 0.0042 redundant alerts per incident and buys 0.0005 recall; moving from 10 s to the deployed 120 s costs 0.029 recall — a factor of eleven more masked incidents — and buys 0.6 percentage points of suppression. Moving to 900 s costs 22.6% of distinct incidents. The curve is flat to the left of the knee in the quantity that matters and steep to the right of it, which is why an estimator that errs low is safer than one that errs high, and why we would rather a deployment take 6.6 s from the Debar–Wespi rule than 120 s from a geometric argument.

Method	Alerts	Reduction	Incident recall	Distinct recall	Alerts/incident
naive (one per detection)	59,984	—	1.000	1.000	5.00
per-modality dedup	15,788	73.7%	1.000	0.9635	1.32
confidence voting	7,130	88.1%	0.365	0.3483	1.63
entity-agnostic collapse	11,626	80.6%	1.000	0.9680	0.97

Table 6: **Collapse strategies (synthetic)**, $W = 120$ s. Confidence voting achieves the largest reduction by discarding 63.5% of incidents entirely. It is reported because it is the obvious alternative, and rejected because no control room can run at 0.365 incident recall.

Dual-tagged share	Per-modality alerts	Entity-agnostic alerts	Extra reduction
0.00	11,654	11,654	0.0%
0.15	12,990	11,633	10.5%
0.30	14,415	11,671	19.0%
0.45	15,715	11,605	26.2%
0.70	17,973	11,637	35.3%
1.00	20,614	11,644	43.5%

Table 7: **Marginal value of crossing modalities (synthetic)**. Recall is 1.000 for both methods at every share. The entity-agnostic column is flat because it is insensitive to how many modalities tagged the incident; that insensitivity is the entire benefit.

7 Evaluation

Table 5 is the paper’s central result. Read it in two parts.

Below the knee, collapse is nearly free. Going from 0 to 15s removes 47,976 of the 47,984 redundant alerts and masks 47 incidents: 80.1% fewer alerts for 0.4% of distinct-incident recall. No deployment should decline this.

Above the knee, the curve is almost flat in the benefit and steep in the cost. From 15s to 120s, alerts fall by a further 335 (2.8% of the remainder, 0.5 percentage points of total reduction) while masked incidents rise from 47 to 384, an eightfold increase. Recall drops from 0.9961 to 0.9680. The deployed value therefore sits well past the knee: it pays 337 masked incidents for a reduction an operator would not notice. At 1,800s the trade is grotesque — 85.0% reduction against 0.7510 recall — but the shape of the curve is already established by 120s.

This coarse sweep has no sample point between 5 and 15s. Section 6 repeats it at 24 points and locates the knee at 8–10s, inside that gap; the recommendation is $W = 10$ s and the numbers in this table should be read against Table 3, which supersedes them.

7.1 Cross-modal collapse against per-modality deduplication

Table 7 isolates the one design decision in the predicate that is genuinely ours to defend: keying on the entity rather than on (entity, modality). The benefit is exactly proportional to how often an incident is seen by more than one analytic, from zero at 0% dual tagging to 43.5% at 100%. Our estate runs at roughly 40% dual tagging (4,859 of 12,000 incidents), placing the real benefit near 24%. A deployment running one analytic per camera should not adopt this and gains nothing from it; that conditional is the honest form of the claim.

8 Related work

Covered in §2 and conceded there. The one comparison we owe is Valdes and Skinner [27], whose weighted-similarity aggregation is the principal alternative to a hard entity key: it fuses

on graded feature agreement rather than exact match. It would handle entity-key errors — a misread plate — which our predicate cannot, and a deployment with noisy keys should prefer it. We do not claim superiority over it; our estate’s keys come from ANPR with a confidence gate, and exact matching is adequate there. Cheng et al. [7] and Ye et al. [30] are the alternatives for graded matching across cameras.

9 Limitations

1. **Synthetic ground truth.** “Distinct incident” is a label our generator emits. In an estate, whether two appearances of one vehicle forty seconds apart are one incident or two is an operator judgement, and the whole curve in Table 5 depends on that judgement. The replication that matters is operator-marked incident boundaries on real footage; if operators mark a longer typical event than our generator, the knee moves right and 120s may be defensible on grounds we did not test. The estimators in §6 are the part of this paper that transfers regardless: they consume an inter-observation-time column and an incident label, both of which a real deployment can produce. The geometric refutation in §4 does not depend on this, since it compares the deployment against its own stated rule.
2. **Anchored semantics.** Under merging semantics all of these numbers change and the masking cost grows without bound.
3. **One estate, one entity-key source.** Key errors are not modelled.
4. **A single global W .** Table 1 shows the geometric quantity varies $7.6\times$ across cameras. Whether a per-camera W pays is not answered here.
5. **Case grouping is a different problem.** An operator may want all alerts for one entity grouped in the interface for hours while still being notified of each. That is a presentation decision and it is arguably the right answer to the fatigue problem; this paper addresses suppression, not grouping.

10 Conclusion

The collapse predicate is not new and we do not claim it. Its window is the parameter that decides whether it helps, and the derivation our deployment used for that window — field-of-view depth over permitted speed — is refuted by the estate’s own camera parameters at all 200 cameras, by a factor of 12 to 94, and is measuring the wrong quantity besides. On a 24-point operating curve the knee is at 8–10s, and four independent estimators agree to within 3.4s; the deployed 120s buys 0.6 percentage points of extra suppression for eleven times the masked incidents. We recommend $W = 10$ s here, and more generally that deployments derive W from the inter-observation-time distribution under an explicit cost weighting, publish the weighting, prefer estimators that read the error curve directly over ones that fit a latent mixture, and stop deriving the window from geometry.

Reproduction

```
cd code
pip install -r requirements.txt
python -m pytest -q
python -m prresearch.p5_fusion.experiment
```

Fixed seeds; `results/p5_fusion.json` is byte-identical across runs. Code and data: <https://github.com/amitduabits/PRAHARI-P5-EventSchema>

References

- [1] Naseeb Ahmed Adnan, Iman Izadi, and Tongwen Chen. On expected detection delays for alarm systems with deadbands and delay-timers. *Journal of Process Control*, 21(9): 1318–1331, 2011. doi: 10.1016/j.jprocont.2011.06.019.
- [2] Muhammad Shahzad Afzal, Tongwen Chen, Ali Bandehkhoda, and Iman Izadi. Analysis and design of time-deadbands for univariate alarm systems. *Control Engineering Practice*, 71:96–107, 2018. doi: 10.1016/j.conengprac.2017.10.016.
- [3] Tyler Akidau, Robert Bradshaw, Craig Chambers, Slava Chernyak, Rafael J. Fernández-Moctezuma, Reuven Lax, Sam McVeety, Daniel Mills, Frances Perry, Eric Schmidt, and Sam Whittle. The dataflow model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-of-order data processing. *Proceedings of the VLDB Endowment*, 8(12):1792–1803, 2015. doi: 10.14778/2824032.2824076.
- [4] Bushra A. Alahmadi, Louise Axon, and Ivan Martinovic. 99% false positives: A qualitative study of SOC analysts’ perspectives on security alarms. In *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, 2022. URL <https://www.usenix.org/conference/usenixsecurity22/presentation/alahmadi>. No DOI assigned.
- [5] Stefan Axelsson. The base-rate fallacy and the difficulty of intrusion detection. *ACM Transactions on Information and System Security*, 3(3):186–205, 2000. doi: 10.1145/357830.357849.
- [6] Lara D. Catledge and James E. Pitkow. Characterizing browsing strategies in the world-wide web. *Computer Networks and ISDN Systems*, 27(6):1065–1073, 1995. doi: 10.1016/0169-7552(95)00043-7.
- [7] Yue Cheng, Iman Izadi, and Tongwen Chen. Pattern matching of alarm flood sequences by a modified Smith–Waterman algorithm. *Chemical Engineering Research and Design*, 91(6):1085–1094, 2013. doi: 10.1016/j.cherd.2012.11.001.
- [8] Gianpaolo Cugola and Alessandro Margara. Processing flows of information: From data stream to complex event processing. *ACM Computing Surveys*, 44(3):15:1–15:62, 2012. doi: 10.1145/2187671.2187677.
- [9] Hervé Debar and Andreas Wespi. Aggregation and correlation of intrusion-detection alerts. In *Recent Advances in Intrusion Detection (RAID 2001)*, volume 2212 of *Lecture Notes in Computer Science*, pages 85–103. Springer, 2001. doi: 10.1007/3-540-45474-8_6.
- [10] Hervé Debar, David A. Curry, and Benjamin S. Feinstein. The intrusion detection message exchange format (IDMEF). Technical Report RFC 4765, IETF, 2007. URL <https://www.rfc-editor.org/rfc/rfc4765.html>. IETF RFC (non-paper reference).
- [11] Ahmed K. Elmagarmid, Panagiotis G. Ipeirotis, and Vassilios S. Verykios. Duplicate record detection: A survey. *IEEE Transactions on Knowledge and Data Engineering*, 19(1):1–16, 2007. doi: 10.1109/TKDE.2007.250581.
- [12] *EEMUA Publication 191: Alarm Systems — A Guide to Design, Management and Procurement*. Engineering Equipment and Materials Users’ Association, London, UK, 3rd edition, 2013. URL <https://www.eemua.org/products/publications/print/eemua-publication-191>. Industry guide (non-paper reference); edition/year should be confirmed against the copy used, as later editions exist.

- [13] Tom Fawcett. An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8):861–874, 2006. doi: 10.1016/j.patrec.2005.10.010.
- [14] Ivan P. Fellegi and Alan B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64(328):1183–1210, 1969. doi: 10.1080/01621459.1969.10501049.
- [15] Aaron Halfaker, Oliver Keyes, Daniel Kluver, Jacob Thebault-Spieker, Tien Nguyen, Kenneth Shores, Anuradha Uduwage, and Morten Warncke-Wang. User session identification based on strong regularities in inter-activity time. In *Proceedings of the 24th International Conference on World Wide Web (WWW '15)*, pages 410–418, 2015. doi: 10.1145/2736277.2741117. arXiv:1411.2878.
- [16] David L. Hall and James Llinas. An introduction to multisensor data fusion. *Proceedings of the IEEE*, 85(1):6–23, 1997. doi: 10.1109/5.554205.
- [17] *ANSI/ISA-18.2-2016: Management of Alarm Systems for the Process Industries*. International Society of Automation, Research Triangle Park, NC, USA, 2016. URL <https://www.isa.org/standards-and-publications/isa-standards>. Standard (non-paper reference).
- [18] Klaus Julisch. Clustering intrusion detection alarms to support root cause analysis. *ACM Transactions on Information and System Security*, 6(4):443–471, 2003. doi: 10.1145/950191.950192.
- [19] Bahador Khaleghi, Alaa Khamis, Fakhreddine O. Karay, and Saiedeh N. Razavi. Multisensor data fusion: A review of the state-of-the-art. *Information Fusion*, 14(1):28–44, 2013. doi: 10.1016/j.inffus.2011.08.001.
- [20] Mark Meiss, John Duncan, Bruno Gonçalves, José J. Ramasco, and Filippo Menczer. What’s in a session: Tracking individual behavior on the web. In *Proceedings of the 20th ACM Conference on Hypertext and Hypermedia (HT '09)*, pages 173–182, 2009. doi: 10.1145/1557914.1557946.
- [21] Peng Ning, Yun Cui, and Douglas S. Reeves. Constructing attack scenarios through correlation of intrusion alerts. In *Proceedings of the 9th ACM Conference on Computer and Communications Security (CCS '02)*, pages 245–254, 2002. doi: 10.1145/586110.586144.
- [22] OCSF Project. Open cybersecurity schema framework (OCSF) schema. Open specification, 2026. URL <https://schema.ocsf.io/>. Non-paper reference; continuously versioned, year is the access year.
- [23] Ville Satopää, Jeannie Albrecht, David Irwin, and Barath Raghavan. Finding a “kneedle” in a haystack: Detecting knee points in system behavior. In *2011 31st International Conference on Distributed Computing Systems Workshops (ICDCSW)*, pages 166–171, 2011. doi: 10.1109/ICDCSW.2011.20.
- [24] Alan N. Steinberg, Christopher L. Bowman, and Franklin E. White. Revisions to the JDL data fusion model. In *Sensor Fusion: Architectures, Algorithms, and Applications III*, volume 3719 of *Proceedings of SPIE*, pages 430–441, 1999. doi: 10.1117/12.341367.
- [25] Sathya Chandran Sundaramurthy, Alexandru G. Bardas, Jacob Case, Xinming Ou, Michael Wesch, John McHugh, and S. Raj Rajagopalan. A human capital model for mitigating security analyst burnout. In *Eleventh Symposium on Usable Privacy and Security (SOUPS 2015)*. USENIX Association, 2015. URL <https://www.usenix.org/conference/soups2015/proceedings/presentation/sundaramurthy>. No DOI assigned.

- [26] Shahroz Tariq, Mohan Baruwal Chhetri, Surya Nepal, and Cecile Paris. Alert fatigue in security operations centres: Research challenges and opportunities. *ACM Computing Surveys*, 57(9):1–38, 2025. doi: 10.1145/3723158.
- [27] Alfonso Valdes and Keith Skinner. Probabilistic alert correlation. In *Recent Advances in Intrusion Detection (RAID 2001)*, volume 2212 of *Lecture Notes in Computer Science*, pages 54–68. Springer, 2001. doi: 10.1007/3-540-45474-8_4.
- [28] Jiandong Wang, Fan Yang, Tongwen Chen, and Sirish L. Shah. An overview of industrial alarm systems: Main causes for alarm overloading, research status, and open problems. *IEEE Transactions on Automation Science and Engineering*, 13(2):1045–1061, 2016. doi: 10.1109/TASE.2015.2464234.
- [29] Jiandong Wang, Zhen Wang, Xuan Zhou, and Fan Yang. Design of delay timers based on estimated probability mass functions of alarm durations. *Journal of Process Control*, 110: 154–165, 2022. doi: 10.1016/j.jprocont.2022.01.002.
- [30] Mang Ye, Jianbing Shen, Gaojie Lin, Tao Xiang, Ling Shao, and Steven C. H. Hoi. Deep learning for person re-identification: A survey and outlook. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(6):2872–2893, 2022. doi: 10.1109/TPAMI.2021.3054775.